JOC The Journal of Organic Chemistry

pubs.acs.org/joc Article

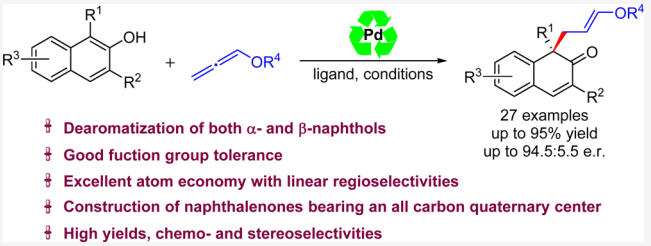
Pd-Catalyzed Dearomative Asymmetric Allylic Alkylation of Naphthols with Alkoxyallenes

Jinjin Hu, Shulei Pan, Shuai Zhu, Peiyuan Yu, Ruigang Xu, Guofu Zhong, and Xiaofei Zeng*

Cite This: J. Org. Chem. 2020, 85, 7896–7904 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: A Pd-catalyzed linear selective intermolecular asymmetric dearomative allylic alkylation reaction of naphthols with alkoxyallenes under mild reaction conditions is reported. The transformation is successfully promoted by $Pd_2(dba)_3$ and the chiral Trost ligand and provides a general atom-efficient protocol to obtain various β-naphthalenones bearing an all carbon quaternary stereogenic center in good yields and chemo- and stereoselectivities.



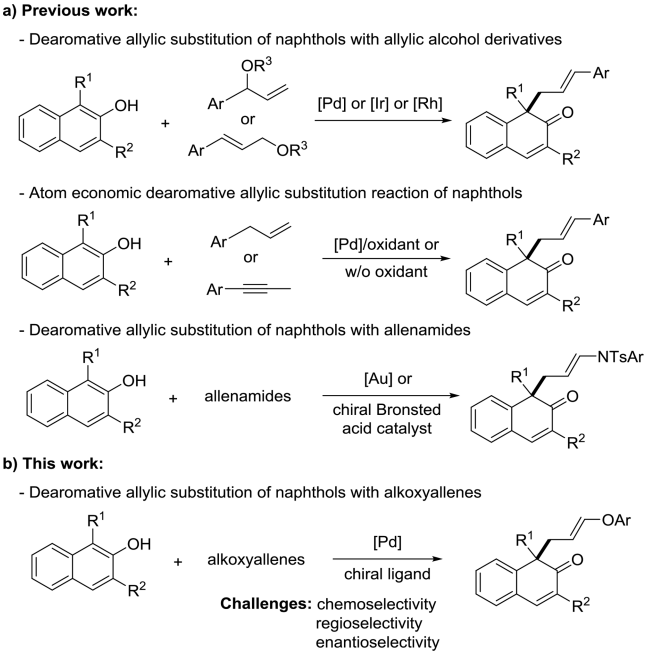
Downloaded via FUDAN UNIV on February 20, 2021 at 04:05:08 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

INTRODUCTION

Cyclohexadienones with ortho-quaternary carbon centers constitute the main structural elements of many natural products[1] and biologically active compounds.[2] Thus, tremendous attention from synthetic chemists has been received for the construction of a cyclohexadienone skeleton.[3] In recent years, the transition-metal-catalyzed asymmetric dearomative allylic substitution reactions[4] have been proven to be one of the most efficient strategies for preparing target molecules from simple phenols or naphthols.[5] A series of successful examples have been realized by using allylic alcohol[6] and their derivatives[7] as the π-allylic precursors (Scheme 1a). To improve the atom economy, Lin and co-workers reported a redox-neutral strategy for the dearomative coupling of β-naphthols with alkynes via Pd hydride catalysis, providing the desired naphthalenones in good yields.[8] Later on, Yang and co-workers disclosed a highly efficient palladium-catalyzed direct oxidative allylic C−H dearomatization of 2-naphthols to access naphthalenones.[9] Despite the impressive advances in dearomative allylic substitution of naphthols, exploring new types of electrophiles to enrich the application scope of these transformations is still challenging and highly desirable. In this regard, Bandini and co-workers reported gold(I)-catalyzed dearomatization of 2-naphthols with allenamides in high yields and selectivities.[10] The asymmetric version of this reaction was then reported by Shao and co-workers by using chiral Brønsted acid catalysis, affording the target allylic dearomatization products in good to excellent enantioselectivities.[11]

With the pioneering reports by Trost and co-workers,[12] alkoxyallenes have been widely employed as useful π-allyl fragment precursors for transition-metal-catalyzed allylic addition reactions to pronucleophiles, and considerable interest has been received from synthetic chemists due to the highly branched regioselectivity and efficient atom economy.[13] Numerous soft[14] and hard[15] nucleophiles have been involved in this reaction during the last decade. In line with our interest in the development of catalytically

Scheme 1. Catalytic Dearomatization Reactions of Naphthols



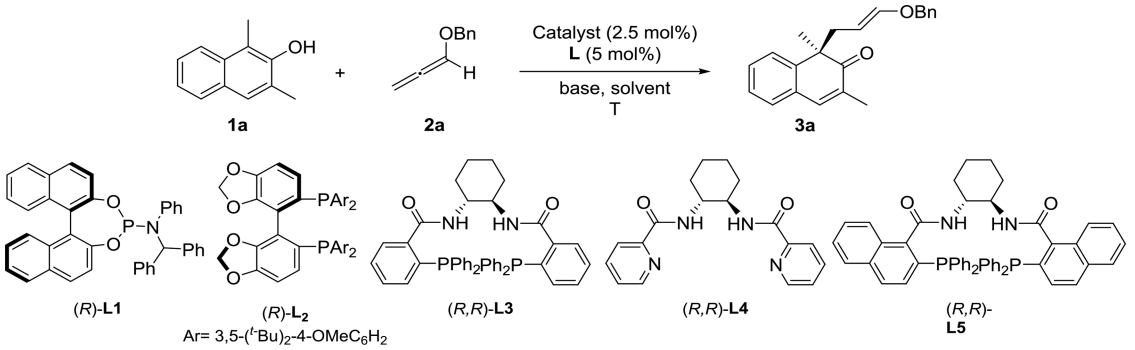
Received: March 4, 2020
Published: May 29, 2020



ACS Publications © 2020 American Chemical Society

https://dx.doi.org/10.1021/acs.joc.0c00582
J. Org. Chem. 2020, 85, 7896–7904

Table 1. Reaction Condition Optimization[a]



entry	L	catalyst	base	solvent	T [°C]	yield [%][b]	e.r.[c]
1	(*R*)-**L1**	$Pd(dba)_2$	DBU	DCE	60	64	55:45
2	(*R*)-**L2**	$Pd(dba)_2$	DBU	DCE	60	36	56:44
3	(*R,R*)-**L3**	$Pd(dba)_2$	DBU	DCE	60	58	78:22
4	(*R,R*)-**L4**	$Pd(dba)_2$	DBU	DCE	60		
5	(*R,R*)-**L5**	$Pd(dba)_2$	DBU	DCE	60	63	82:18
6	(*R,R*)-**L5**	$Pd_2(dba)_3$	DBU	DCE	60	66	89:11
7	(*R,R*)-**L5**	$Pd(OAc)_2$	DBU	DCE	60	61	77:23
8	(*R,R*)-**L5**	$Pd_2(dba)_3$	DBU	THF	60	25	91:9
9	(*R,R*)-**L5**	$Pd_2(dba)_3$	Et_3N	THF	60	35	60:40
10	(*R,R*)-**L5**	$Pd_2(dba)_3$	TMG	THF	60	34	91.5:8.5
11	(*R,R*)-**L5**	$Pd_2(dba)_3$	TMG[d]	PO[e]	60	81	92:8
12	(*R,R*)-**L5**	$Pd_2(dba)_3$	TMG	PO	25		

[a]General conditions: **1a** (0.2 mmol), **2a** (0.3 mmol), $Pd_2(dba)_3$ (2.5 mol %), ligand (5 mol %), and base (0.2 mmol) in 2 mL of solvent under argon at 60 °C for 12 h. [b]Yield of the isolated product. [c]Determined by HPLC on a chiral stationary phase. [d]TMG = tetramethylguanidine. [e]PO = propylene oxide.

asymmetric dearomatic reactions of naphthols[6b] and selectivity-controllable allylic substitution reactions,[16] we assumed that an asymmetric metal-catalyzed intermolecular nucleophilic addition reaction to alkoxyallenes involving π-allyl intermediates would be highly suitable for β-naphthols, allowing access to dearomative cyclohexadienone derivatives. Herein, we report the first Pd-catalyzed linear selective asymmetric allylic dearomatization of β-naphthols with alkoxyallenes for the construction of β-naphthalenones bearing an all-carbon quaternary chiral center in good yields and enantioselectivities. Interestingly, α-naphthols were also found to be suitable substrates in this reaction system.

RESULTS AND DISCUSSION

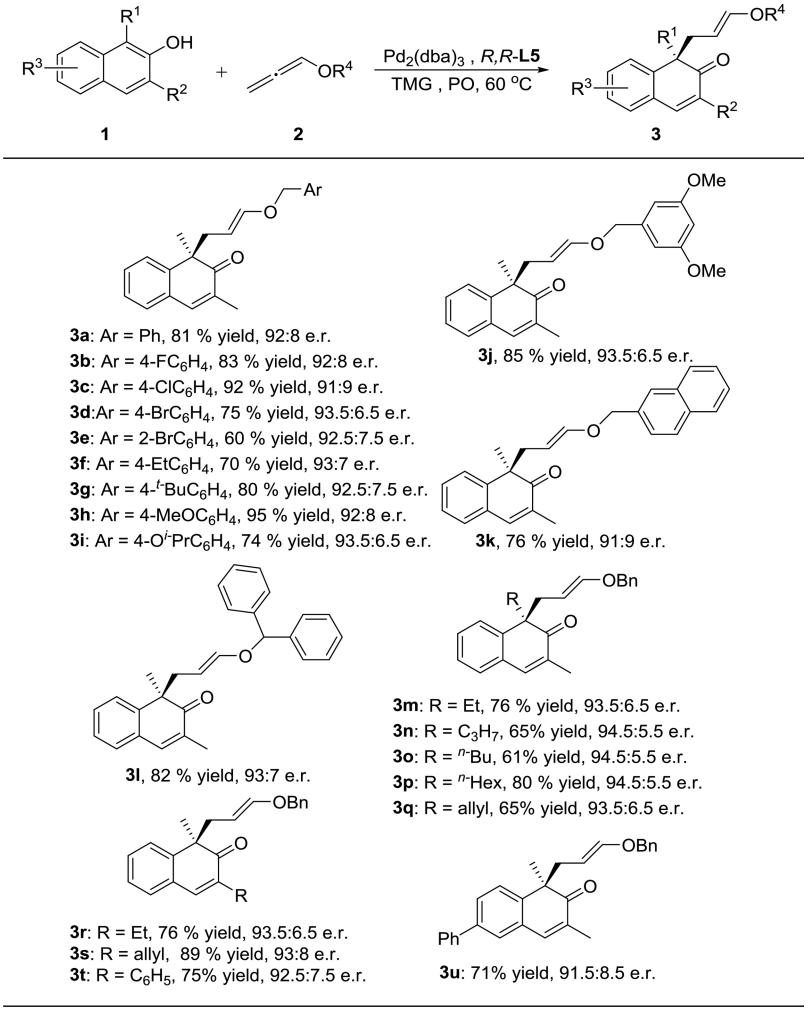
We initiated our study by using 1,3-$(Me)_2$-naphthyl-2-ol (**1a**) and alkoxyallenes (**2a**) as the model substrates to optimize the reaction conditions. The initial test was performed in the presence of 2.5 mol % $Pd(dba)_2$, 5 mol % ligand (*R*)-**L1**[17] as the catalyst, 1.0 equiv of DBU as the base, and dicholoroethane (DCE) as solvent, and the reaction proceeded smoothly with high conversion of **1a**. Interestingly, a linear selective product (**3a**) was formed in a 64% yield and 55:45 e.r. (Table 1, entry 1). Encouraged by this positive result, we then screened a series of ligands, and the Trost ligand (*R,R*)-**L5** proved to be the best with respect to enantioselectivities and yields for the model reaction (Table 1, entry 5). Taking (*R,R*)-**L5** as the optimal ligand, different palladium precursors such as $Pd_2(dba)_3$, $Pd_2(dba)_3CHCl_3$, and $Pd(OAc)_2$ were tested, and $Pd_2(dba)_3$ was found to be the best, furnishing the corresponding product in a 66% yield and 89:11 e.r. (Table 1, entry 6). On further optimizing the reaction conditions by varying the solvent, base, and reaction temperature (for details, see the Supporting Information, Table S1), we identified the following optimal conditions: **1a** (0.2 mmol) was treated with **2a** (1.5 equiv) in the presence of $Pd_2(dba)_3$ (2.5 mol %), (*R,R*)-**L5** (5 mol %), and DBU (1.0 equiv) in propylene oxide (PO, 0.1 M) at 60 °C for 12 h, and **3a** (E/Z > 19:1) was obtained in an 81% yield with 92:8 e.r. (Table 1, entry 11).

With the optimized reaction conditions in hand, we investigated the substrate scope of this dearomative asymmetric allylic alkylation process (Scheme 2). Alkoxyallenes bearing different electron-withdrawing or electron-donating groups on the phenyl ring underwent the reaction smoothly, delivering the corresponding β-naphthalenones in high yields and good stereoselectivities. (**3b**−**3l**, 60−92% yields, 91:9−93.5:6.5 e.r.'s). Gratifyingly, β-naphthols with different substitution groups at 1- or 3-positions, such as allyl, alkyl, and methyl groups, were also well tolerated in the reaction, giving **3m**−**3t** in good yields and enantioselectivities. (61−89% yield, 92.5:7.5−94.5:5.5 e.r.'s). Furthermore, 1,3-dimethyl-6-phenylnaphthalen-2-ol (**1u**) was also a good substrate, providing the desired dearomative product **3u** with a good yield and enantioselectivity. To confirm the absolute configuration (AC) of the products, ECD spectra were calculated by the TD-DFT method, which has been proven to be useful in predicting ECD spectra and assigning the AC of organic molecules. The *R* configuration could be reliably assigned to compound **3a** (see Figure S1 in the Supporting Information).

Next, we examined the compatibility of our reaction protocol with α-naphthols under standard conditions. As shown in Scheme 3, 2-allylnaphth-1-ols with different substitution groups react with different alkoxyallenes smoothly, giving the desired chiral α-naphthalenones (**5a**−**5f**) with good

https://dx.doi.org/10.1021/acs.joc.0c00582
J. Org. Chem. 2020, 85, 7896−7904

Scheme 2. Scope of β-Naphthol and Alkoxyallenes[a]



[a]Reaction conditions: **1** (0.2 mmol), **2** (0.3 mmol), $Pd_2(dba)_3$ (2.5 mol %), ligand (5 mol %), and base (0.2 mmol) in 2 mL of solvent at 60 °C overnight. Yield of the isolated product. The e.r. value was determined by chiral HPLC analysis.

yields (up to a 79% yield) and enantioselectivities (up to 94:6 e.r.). To demonstrate the utility of the Pd-catalyzed asymmetric dearomatization reaction of naphthols, preparative-scale synthesis of **3a** was carried out to efficiently afford 1.81 g of **3a** in a 78% yield and 92:8 e.r. (Scheme 4a). There was almost no change in either the yield or enantioselectivity, suggesting that large-scale chemical production of chiral naphthalenones may be feasible. Diastereoselective reduction of the ketone group of **3a** with $NaBH_4$ gave alcohol **3ab** in a 95% yield with 95:5 d.r. and 91.5:8.5 e.r. (Scheme 4b).

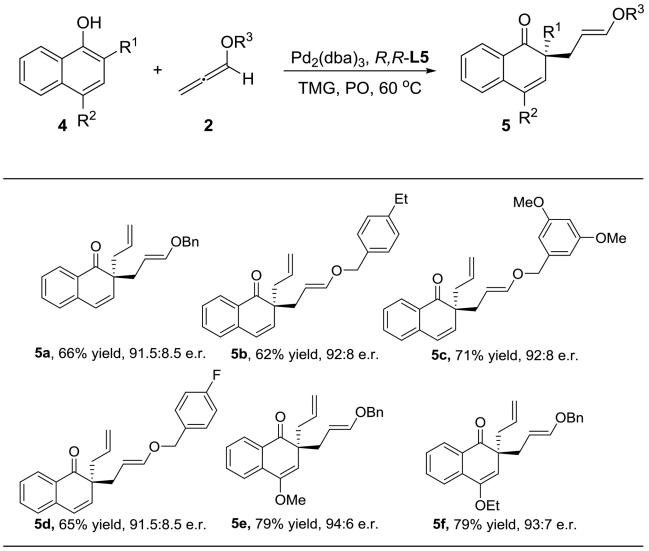
To gain insight into the reaction mechanism, we performed several control experiments. First, the dearomatization reaction was carried out under standard conditions without the base, and it was found that both yields and enantioselectivities were decreased (31% yield and 60:40 e.r.) (Scheme 4c). This outcome indicated that the **L***Pd(II)−H species **II** generated from the chiral palladium complex with naphthol (**1a**) could react with allene (**2a**) automatically without a base and, however, the reaction is too slow, which resulted in a low yield. In addition, 2-methoxy-1,3-dimethylnaphthalene (MeO-**1a**) could not proceed in the reaction at all (Scheme 4d). Then, a deuterium-labeling experiment was carried out by adding 1 drop of D_2O in the reaction under standard conditions. It could be seen that 61% of the H atoms at the middle carbon of allene was deuterated. (Scheme 4e) Based on these experimental results and previous related work,[18] a possible catalytical cycle was proposed. First, an oxidative addition of **L***Pd(0) **I**, *in situ* generated from $Pd_2(dba)_3$ with a chiral ligand (**L***), to the naphthol (**1a**) generates **L***Pd(II)−H species **II**, which can efficiently prohibit the *O*-position from attacking the allene moiety.[15c] Then, the hydropalladation of **II** with allene (**2a**) forms a transition state **III**, and **III** will facilitate the nucleophilic addition of the naphthol to the π-allyl fragment, allowing access to the desired linear product enantioselectively through a reductive elimination process (Scheme 5).

CONCLUSIONS

In summary, we have developed a Pd-catalyzed dearomative asymmetric allylic alkylation reaction of naphthols with alkoxyallenes. Both α- and β-naphthols were well tolerated in the reaction and provided various α- and β-naphthalenones bearing an all carbon quaternary center in good yields (up to 95%) and enantioselectivities (up to 94.5:5.5 e.r.) with good functional-group tolerance. The catalytic process involved the hydropalladation of alkoxyallene and subsequent allyl sub-

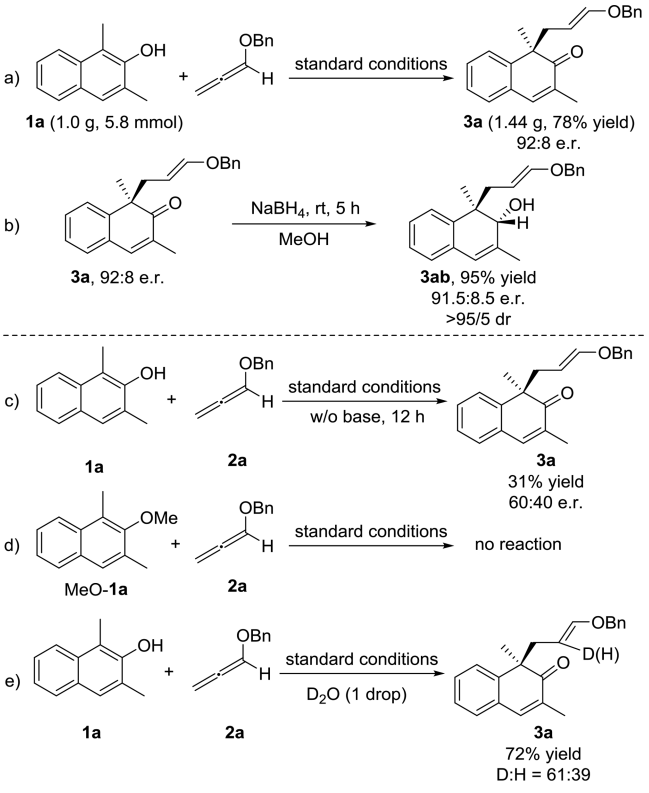
https://dx.doi.org/10.1021/acs.joc.0c00582
J. Org. Chem. 2020, 85, 7896−7904

Scheme 3. Scope of α-Naphthol and Alkoxyallenes[a]



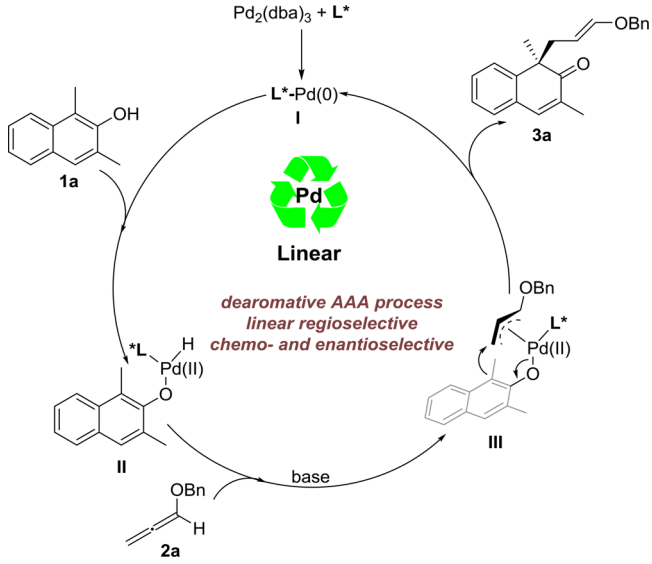
[a]Reaction conditions: **1** (0.2 mmol), **2** (0.3 mmol), $Pd_2(dba)_3$ (2.5 mol %), ligand (5 mol %), and base (0.2 mmol) in 2 mL of solvent at 60 °C overnight. Yield of the isolated product. The e.r. value was determined by chiral HPLC analysis.

Scheme 4. Chemical Manipulation of the Dearomatized Naphthalenone Compounds (3) and the Mechanism Proposal



stitution using naphthol as a pronucleophile, affording the desired products in absolutely linear selectivity. This approach provides a concise route for access to enantiomeric enriched naphthalenones.

Scheme 5. Proposed Catalytical Cycle



EXPERIMENTAL SECTION

General Information. Analytical thin layer chromatography (TLC) was performed using a Merck 60F254 precoated silica gel plate (0.2 mm in thickness). Subsequent to elution, plates were visualized using UV radiation (254 nm) on Spectroline Model ENF-24061/F of 254 nm. Further visualization was possible by staining with a basic solution of potassium permanganate or acidic solution of ceric molybdate. Flash column chromatography was performed using Merck aluminum oxide90 active, neutral with freshly distilled solvents. Columns were typically packed as slurry and equilibrated with the appropriate solvent system prior to use. Proton nuclear magnetic resonance spectra (^{1}H NMR) were recorded on a Bruker AMX 500 spectrophotometer ($CDCl_3$ as solvent). Chemical shifts for ^{1}H NMR spectra are reported as δ in units of parts per million (ppm) downfield from $SiMe_4$ (0.0) and relative to the signal of chloroform-*d* (7.26, singlet). Multiplicities were given as s (singlet), d (doublet), t (triplet), dd (doublets of doublet), or m (multiplets). The number of protons (*n*) for a given resonance is indicated by *n*H. Coupling constants are reported as a *J* value in hertz. Carbon nuclear magnetic resonance spectra (^{13}C NMR) are reported as δ in units of parts per million (ppm) downfield from $SiMe_4$ (0.0) and relative to the signal of chloroform-*d* (77.0, triplet). Enantiomeric excesses were determined by high-performance liquid chromatography (HPLC) analysis on a chiral stationary phase, CHIRALCEL IC, CHIRALCEL OD, or CHIRALCEL AD. Optical rotations were measured in $CHCl_3$ on a Schmidt + Haensdchpolarimeter (Polartronic MH8) with a 10 cm cell (*c* given in g/100 mL). High-resolution mass spectrometry (HRMS) was recorded on a QTOF Premier for ESI+. The racemic products used to determine the ee values were synthesized by using $Pd(PPh_3)_4$ as the catalyst. The starting materials **1** and **2** were prepared according to the literature reported.[19]

General Procedure for the Dearomative Asymmetric Allylic Reaction of β-Naphthols with Alkoxyallenes. To a 5 mL sealed tube was added β-naphthols (**1**, 0.2 mmol, 1.0 equiv) and alkoxyallenes (**2**, 0.3 mmol, 1.5 equiv) in 2 mL of PO followed by addition of $Pd_2(dba)_3$ (2.5 mol %), TMG (0.2 mmol), and (*R*,*R*)-**L5** (5 mol %), and the reaction was stirred with noble gas protection at 60 °C in an oil bath until completion of the reaction (monitored by TLC). The reaction mixture was concentrated under reduced pressure, and the resulting residue was purified by flash chromatography (EtOAc/hexane) to provide **3**.

General Procedure for the Dearomative Asymmetric Allylic Reaction of α-Naphthols with Alkoxyallenes. To a 5 mL sealed tube was added α-naphthols (**4**, 0.2 mmol, 1.0 equiv) and alkoxyallenes (**2**, 0.3 mmol, 1.5 equiv) in 2 mL of PO followed by addition of $Pd_2(dba)_3$ (2.5 mol %), TMG (0.2 mmol), and (*R*,*R*)-**L5**

https://dx.doi.org/10.1021/acs.joc.0c00582
J. Org. Chem. 2020, 85, 7896−7904

(5 mol %), and the reaction was stirred with noble gas protection at 60 °C in an oil bath until completion of the reaction (monitored by TLC). The reaction mixture was concentrated under reduced pressure, and the resulting residue was purified by flash chromatography (EtOAc/hexane) to provide **6**.

Procedure for the Synthesis of Alcohol 6a. To a solution of **3a** (31.8 mg, 0.1 mmol, 92:8 e.r.) in anhydrous MeOH (2.0 mL), $NaBH_4$ (4.9 mg, 0.12 mmol) was added. The reaction mixture was stirred at rt for 5 h. After the reaction was complete (monitored by TLC), the solvents were removed under reduced pressure. Then, the diastereomeric ratio was determined by ^1H NMR of the crude reaction mixture. Then, the residue was purified by silica gel column chromatography (PE/EtOAc = 9/1) to afford the desired product **3ab**.

(*R*,*E*)-1-(3-(Benzyloxy)allyl)-1,3-dimethylnaphthalen-2(1*H*)-one (3a). Colorless oil, (petroleum ether/EtOAc = 40:1−10:1); 51.5 mg, yield (81%); $[\alpha]_D^{20}$ = −55.63 (*c* 0.4, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 7.32−7.22 (m, 6H), 7.20 (d, *J* = 7.4 Hz, 1H), 7.17−7.15 (m, 4.4 Hz, 3H), 6.07 (d, *J* = 12.5 Hz, 1H), 4.51 (s, 2H), 4.32 (ddd, *J* = 12.5, 8.3, 7.3 Hz, 1H), 2.62 (dd, *J* = 13.5, 8.5 Hz, 1H), 2.38−2.34 (m, 1H), 1.93 (d, *J* = 1.1 Hz, 3H), 1.43 (s, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 204.26, 147.9, 145.1, 141.7, 137.1, 132.5, 130.4, 128.7, 128.5, 128.4, 127.7, 127.2, 126.6, 126.4, 99.4, 70.9, 51.8, 42.1, 25.9, 15.9. 92:8 e.r. as determined by HPLC (Chiralcel OD, 95:5 hexanes/*i*-PrOH, 1 mL/min), t_r (major) = 9.41 min, t_r (minor) = 8.72 min. HRMS (ESI+) *m*/*z*: $[M + H]^+$ calcd for $C_{22}H_{23}O_2$ 319.1693, found 319.169.

(*R*,*E*)-1-(3-((4-Fluorobenzyl)oxy)allyl)-1,3-dimethylnaphthalen-2(1*H*)-one (3b). Colorless oil, (petroleum ether/EtOAc = 40:1−10:1); 55.9 mg, yield (83%); $[\alpha]_D^{20}$ = −48.85 (*c* 0.4, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 7.33 (d, *J* = 3.7 Hz, 2H), 7.26−7.20 (m, 2H), 7.16 (d, *J* = 0.9 Hz, 1H), 7.12 (dd, *J* = 8.5, 5.5 Hz, 2H), 6.97 (t, *J* = 8.7 Hz, 2H), 6.05 (d, *J* = 12.6 Hz, 1H), 4.47 (d, *J* = 1.3 Hz, 2H), 4.29 (ddd, *J* = 12.5, 8.4, 7.2 Hz, 1H), 2.64 (dd, *J* = 13.6, 8.5 Hz, 1H), 2.39−2.36 (m, 1H), 1.93 (d, *J* = 1.1 Hz, 3H), 1.42 (s, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 204.2, 162.3 (d, *J* = 244.3 Hz), 147.6, 145.1, 141.7, 132.9 (d, *J* = 3.2 Hz), 132.5, 130.3, 129.0 (d, *J* = 12.6 Hz), 128.8, 128.5, 126.6, 126.3, 115.3 (d, *J* = 21.4 Hz), 99.7, 70.2, 51.8, 41.8, 26.3, 15.9. 92:8 e.r. as determined by HPLC (Chiralcel OD, 99:1 hexanes/*i*-PrOH, 1 mL/min), t_r (major) = 15.86 min, t_r (minor) = 17.91 min. HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{22}H_{22}FO_2$ 337.1598, found 337.1589.

(*R*,*E*)-1-(3-((4-Chlorobenzyl)oxy)allyl)-1,3-dimethylnaphthalen-2(1*H*)-one (3c). Colorless oil, (petroleum ether/EtOAc = 40:1−10:1); 64.9 mg, yield (92%); $[\alpha]_D^{20}$ = −28.35 (*c* 0.4, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 7.33−7.31 (m, 2H), 7.26−7.23 (m, 3H), 7.20 (d, *J* = 7.5 Hz, 1H), 7.13 (d, *J* = 1.1 Hz, 1H), 7.06 (d, *J* = 8.4 Hz, 2H), 6.03 (d, *J* = 12.6 Hz, 1H), 4.51−4.45 (m, 2H), 4.25 (ddd, *J* = 12.6, 8.4, 7.2 Hz, 1H), 2.64 (dd, *J* = 13.7, 8.4 Hz, 1H), 2.36 (ddd, *J* = 13.5, 7.1, 0.9 Hz, 1H), 1.92 (d, *J* = 1.2 Hz, 3H), 1.41 (s, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 204.2, 147.5, 145.1, 141.7, 135.7, 133.4, 132.4, 130.3, 128.3, 128.6, 128.5, 128.4, 126.6, 126.3, 99.9, 69.9, 51.7, 41.8, 26.5, 15.8. 91:9 e.r. as determined by HPLC (Chiralcel AD, 99:1 hexanes/*i*-PrOH, 1 mL/min), t_r (major) = 20.25 min, t_r (minor) = 16.70 min. HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{22}H_{22}ClO_2$ 353.1303, found 353.1311.

(*R*,*E*)-1-(3-((4-Bromobenzyl)oxy)allyl)-1,3-dimethylnaphthalen-2(1*H*)-one (3d). Colorless oil, (petroleum ether/EtOAc = 40:1−10:1); 59.6 mg, yield (75%); $[\alpha]_D^{20}$ = −19.98 (*c* 0.4, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 7.39 (d, *J* = 8.4 Hz, 2H), 7.33−7.31 (m, 2H), 7.25−7.23 (m, 1H), 7.20 (d, *J* = 7.2 Hz, 1H), 7.13 (d, *J* = 1.2 Hz, 1H), 6.99 (d, *J* = 8.4 Hz, 2H), 6.03 (d, *J* = 12.6 Hz, 1H), 4.47 (d, *J* = 5.0 Hz, 2H), 4.24 (ddd, *J* = 12.6, 8.4, 7.2 Hz, 1H), 2.64 (dd, *J* = 13.5, 8.5 Hz, 1H), 2.36 (ddd, *J* = 13.5, 7.1, 1.1 Hz, 1H), 1.92 (d, *J* = 1.3 Hz, 3H), 1.41 (s, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 204.1, 147.4, 145.0, 141.7, 136.2, 132.4, 131.5, 130.3, 128.7, 128.7, 128.5, 126.6, 126.3, 121.4, 99.9, 69.9, 51.7, 41.8, 26.5, 15.8. 93.5:6.5 e.r. as determined by HPLC (Chiralcel AD, 95:5 hexanes/*i*-PrOH, 1 mL/min), t_r (major) = 9.60 min, t_r (minor) = 8.41 min. HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{22}H_{22}BrO_2$ 397.0798, found 397.0795.

(*R*,*E*)-1-(3-((2-Bromobenzyl)oxy)allyl)-1,3-dimethylnaphthalen-2(1*H*)-one (3e). Colorless oil, (petroleum ether/EtOAc = 40:1−10:1); 47.6 mg, yield (60%); $[\alpha]_D^{20}$ = −17.55 (*c* 0.4, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 7.50−7.48 (m, 1H), 7.33−7.32 (m, 2H), 7.26−7.19 (m, 4H), 7.16−7.11 (m, 2H), 6.05 (d, *J* = 12.5 Hz, 1H), 4.58 (s, 2H), 4.34 (ddd, *J* = 12.5, 8.3, 7.4 Hz, 1H), 2.63 (ddd, *J* = 13.4, 8.4, 0.7 Hz, 1H), 2.36 (ddd, *J* = 13.5, 7.3, 1.0 Hz, 1H), 1.94 (d, *J* = 1.3 Hz, 3H), 1.44 (s, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 204.3, 147.6, 145.1, 141.7, 136.4, 132.6, 132.5, 130.4, 129.0, 128.8, 128.7, 128.5, 127.4, 126.6, 126.4, 122.3, 99.8, 70.3, 51.8, 42.2, 25.9, 15.9. 92.5:7.5 e.r. as determined by HPLC (Chiralcel IC, 90:10 hexanes/*i*-PrOH, 1 mL/min), t_r (major) = 8.10 min, t_r (minor) = 9.52 min. HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{22}H_{22}BrO_2$ 397.0798, found 397.0801.

(*R*,*E*)-1-(3-((4-Ethylbenzyl)oxy)allyl)-1,3-dimethylnaphthalen-2(1*H*)-one (3f). Colorless oil, (petroleum ether/EtOAc = 40:1−10:1); 48.6 mg, yield (70%); $[\alpha]_D^{20}$ = −57.42 (*c* 0.4, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 7.32 (d, *J* = 3.7 Hz, 2H), 7.25−7.19 (m, 2H), 7.15−7.12 (m, 3H), 7.09 (d, *J* = 8.1 Hz, 2H), 6.06 (d, *J* = 12.5 Hz, 1H), 4.47 (s, 2H), 4.31 (ddd, *J* = 12.5, 8.3, 7.3 Hz, 1H), 2.66−2.59 (m, 3H), 2.36 (ddd, *J* = 13.5, 7.1, 0.7 Hz, 1H), 1.93 (d, *J* = 1.1 Hz, 3H), 1.42 (s, 3H), 1.22 (t, *J* = 7.6 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ ^{13}C NMR (125 MHz, $CDCl_3$) δ 204.3, 148.0, 145.1, 143.8, 141.65, 134.3, 132.5, 130.4, 128.7, 128.5, 127.9, 127.5, 126.6, 126.4, 99.2, 70.9, 51.8, 42.2, 28.6, 25.9, 15.9, 15.7. 93:7 e.r. as determined by HPLC (Chiralcel AD, 99:1 hexanes/*i*-PrOH, 1 mL/min), t_r (major) = 15.40 min, t_r (minor) = 14.13 min. HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{24}H_{27}O_2$ 347.2006, found 347.2001.

(*R*,*E*)-1-(3-((4-(*tert*-Butyl)benzyl)oxy)allyl)-1,3-dimethylnaphthalen-2(1*H*)-one (3g). Colorless oil, (petroleum ether/EtOAc = 40:1−10:1); 60.0 mg, yield (80%); $[\alpha]_D^{20}$ = −34.92 (*c* 0.4, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 7.33−7.31 (m, 4H), 7.25−7.19 (m, 2H), 7.15 (d, *J* = 1.2 Hz, 1H), 7.11 (d, *J* = 8.3 Hz, 2H), 6.06 (d, *J* = 12.5 Hz, 1H), 4.48 (s, 2H), 4.33 (ddd, *J* = 12.5, 8.4, 7.2 Hz, 1H), 2.61 (dd, *J* = 13.5, 8.5 Hz, 1H), 2.36 (ddd, *J* = 13.5, 7.1, 1.0 Hz, 1H), 1.93 (d, *J* = 1.3 Hz, 3H), 1.43 (s, 3H), 1.31 (s, 9H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 204.3, 150.7, 148.0, 145.1, 141.6, 134.1, 132.6, 130.4, 128.7, 128.5, 127.2, 126.6, 126.4, 125.4, 99.1, 70.9, 51.8, 42.3, 34.6, 31.4, 25.8, 15.9. 92.5:7.5 e.r. as determined by HPLC (Chiralcel OD, 99:1 hexanes/*i*-PrOH, 1 mL/min), t_r (major) = 11.15 min, t_r (minor) = 13.09 min. HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{26}H_{31}O_2$ 375.2319, found 375.233.

(*R*,*E*)-1-(3-((4-Methoxybenzyl)oxy)allyl)-1,3-dimethylnaphthalen-2(1*H*)-one (3h). Colorless oil, (petroleum ether/EtOAc = 40:1−10:1); 66.3 mg, yield (95%); $[\alpha]_D^{20}$ = −73.48 (*c* 0.4, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 7.32 (d, *J* = 3.5 Hz, 2H), 7.25−7.20 (m, 2H), 7.17 (d, *J* = 1.3 Hz, 1H), 7.09 (d, *J* = 8.7 Hz, 2H), 6.82 (d, *J* = 8.7 Hz, 2H), 6.05 (d, *J* = 12.5 Hz, 1H), 4.43 (s, 2H), 4.31 (ddd, *J* = 12.5, 8.5, 7.1 Hz, 1H), 3.79 (s, 3H), 2.62 (ddd, *J* = 13.5, 8.5, 0.7 Hz, 1H), 2.36 (ddd, *J* = 13.5, 7.1, 1.1 Hz, 1H), 1.94 (d, *J* = 1.3 Hz, 3H), 1.42 (s, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 204.3, 159.3, 147.9, 145.2, 141.7, 132.5, 130.4, 129.2, 128.9, 128.7, 128.5, 126.6, 126.4, 113.8, 99.3, 70.8, 55.3, 51.8, 42.1, 26.0, 15.9. 92:8 e.r. as determined by HPLC (Chiralcel OD, 99:1 hexanes/*i*-PrOH, 1 mL/min), t_r (major) = 28.29 min, t_r (minor) = 26.00 min. HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{23}H_{25}O_3$ 349.1798, found 349.1803.

(*R*,*E*)-1-(3-((4-Isopropoxybenzyl)oxy)allyl)-1,3-dimethylnaphthalen-2(1*H*)-one (3i). Colorless oil, (petroleum ether/EtOAc = 40:1−10:1); 55.8 mg, yield (74%); $[\alpha]_D^{20}$ = −48.78 (*c* 0.4, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 7.32 (d, *J* = 3.6 Hz, 2H), 7.25−7.22 (m, 2H), 7.18 (d, *J* = 0.9 Hz, 1H), 7.08 (d, *J* = 8.5 Hz, 2H), 6.81 (d, *J* = 8.6 Hz, 2H), 6.06 (d, *J* = 12.5 Hz, 1H), 4.52 (dt, *J* = 12.1, 6.1 Hz, 1H), 4.42 (s, 2H), 4.31 (ddd, *J* = 12.5, 8.3, 7.3 Hz, 1H), 2.62 (dd, *J* = 13.6, 8.6 Hz, 1H), 2.36 (dd, *J* = 13.5, 7.1 Hz, 1H), 1.94 (d, *J* = 1.0 Hz, 3H), 1.43 (s, 3H), 1.32 (d, *J* = 6.1 Hz, 6H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 204.3, 157.6, 148.0, 145.1, 141.7, 132.5, 130.4, 129.0, 128.9, 128.7, 128.5, 126.6, 126.4, 115.8, 99.2, 70.8, 69.9, 51.8, 42.1, 26.0, 22.1, 15.9. 93.5:6.5 e.r. as determined by HPLC (Chiralcel IC, 95:5 hexanes/*i*-PrOH, 0.5 mL/min), t_r (major) = 42.96 min, t_r

https://dx.doi.org/10.1021/acs.joc.0c00582
J. Org. Chem. 2020, 85, 7896−7904

(minor) = 45.81 min. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{25}H_{29}O_3$ 377.2111, found 377.2096.

(*R*,*E*)-1-(3-((3,5-Dimethoxybenzyl)oxy)allyl)-1,3-dimethylnaphthalen-2(1*H*)-one (3j). Colorless oil, (petroleum ether/EtOAc = 40:1−10:1); 64.4 mg, yield (85%); $[\alpha]_D^{20}$ = −44.50 (c 0.4, $CHCl_3$); ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.32−7.30 (m, 2H), 7.24−7.21 (m, 1H), 7.18 (d, J = 7.4 Hz, 1H), 7.12 (d, J = 1.2 Hz, 1H), 6.36 (t, J = 2.3 Hz, 1H), 6.31 (d, J = 2.3 Hz, 2H), 6.04 (d, J = 12.6 Hz, 1H), 4.45 (s, 2H), 4.29 (ddd, J = 12.5, 8.4, 7.2 Hz, 1H), 3.75 (s, 6H), 2.61 (ddd, J = 13.5, 8.5, 0.6 Hz, 1H), 2.35 (ddd, J = 13.5, 7.1, 1.1 Hz, 1H), 1.91 (d, J = 1.3 Hz, 3H), 1.42 (s, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 204.3, 160.9, 147.6, 145.1, 141.7, 139.6, 132.5, 130.4, 128.7, 128.5, 126.5, 126.3, 104.8, 99.9, 99.5, 70.7, 55.3, 51.7, 42.2, 26.0, 15.8. 93.5:6.5 e.r. as determined by HPLC (Chiralcel IA, 90:10 hexanes/i-PrOH, 1 mL/min), t_r (major) = 7.36 min, t_r (minor) = 6.92 min. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{24}H_{27}O_4$ 379.1904, found 379.1897.

(*R*,*E*)-1,3-Dimethyl-1-(3-(naphthalen-2-ylmethoxy)allyl)-naphthalen-2(1*H*)-one (3k). Colorless oil, (petroleum ether/EtOAc = 40:1−10:1); 56.1 mg, yield (76%); $[\alpha]_D^{20}$ = +0.73 (c 0.4, $CHCl_3$); ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.74 (d, J = 8.2 Hz, 1H), 7.70−7.66 (m, 2H), 7.45−7.43 (m , 3H), 7.37−7.33 (s, 2H), 7.29 (d, J = 8.4 Hz, 2H), 6.97 (dd, J = 8.9, 2.5 Hz, 1H), 6.86 (d, J = 2.3 Hz, 1H), 6.24 (d, J = 12.1 Hz, 1H), 4.93 (dt, J = 12.1, 8.0 Hz, 1H), 2.88 (dd, J = 13.5, 8.4 Hz, 1H), 2.58−2.53 (m, 1H), 2.02 (d, J = 1.1 Hz, 3H), 1.52 (s, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 204.0, 155.0, 144.9, 143.6, 142.1, 134.2, 132.6, 130.5, 129.7, 129.5, 129.0, 128.8, 127.7, 126.9, 126.9, 126.5, 124.3, 118.4, 110.2, 108.6, 51.6, 41.1, 29.7, 27.0, 15.9. 91:9 e.r. as determined by HPLC (Chiralcel AD, 90:10 hexanes/i-PrOH, 1 mL/min), t_r (major) = 8.03 min, t_r (minor) = 6.92 min. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{26}H_{25}O_2$ 369.1849, found 369.1855.

(*R*,*E*)-1-(3-(Benzhydryloxy)allyl)-1,3-dimethylnaphthalen-2(1*H*)-one (3l). Colorless oil, (petroleum ether/EtOAc = 40:1−10:1); 64.8 mg, yield (82%); $[\alpha]_D^{20}$ = −23.08 (c 0.4, $CHCl_3$); ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.20−7.12 (m, 9H), 7.09−7.05 (m, 5H), 6.93 (d, J = 1.1 Hz, 1H), 5.94 (d, J = 12.5 Hz, 1H), 5.44 (s, 1H), 4.28 (ddd, J = 12.5, 8.4, 7.4 Hz, 1H), 2.47 (dd, J = 13.5, 8.5 Hz, 1H), 2.21 (ddd, J = 13.5, 7.3, 1.0 Hz, 1H), 1.77 (d, J = 1.3 Hz, 3H), 1.29 (s, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 204.2, 146.9, 145.0, 141.8, 141.1, 141.1, 132.3, 130.3, 128.6, 128.5, 128.4, 127.6, 127.6, 126.8, 126.7, 126.4, 126.3, 101.3, 82.8, 51.6, 42.4, 25.7, 15.8. 93:7 e.r. as determined by HPLC (Chiralcel IC, 90:10 hexanes/i-PrOH, 1 mL/min), t_r (major) = 8.24 min, t_r (minor) = 9.99 min. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{28}H_{27}O_2$ 395.2006, found 395.2016.

(*R*,*E*)-1-(3-(Benzyloxy)allyl)-1-ethyl-3-methylnaphthalen-2(1*H*)-one (3m). Colorless oil, (petroleum ether/EtOAc = 40:1−10:1); 50.6 mg, yield (76%); $[\alpha]_D^{20}$ = −0.55 (c 0.4, $CHCl_3$); ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.35−7.23 (m, 6H), 7.20 (d, J = 6.8 Hz, 1H), 7.15−7.14 (m, 3H), 6.05 (d, J = 12.5 Hz, 1H), 4.48 (s, 2H), 4.26 (ddd, J = 12.5, 8.5, 7.1 Hz, 1H), 2.64 (dd, J = 13.4, 8.7 Hz, 1H), 2.36 (dd, J = 13.2, 6.8 Hz, 1H), 2.23 (dq, J = 14.6, 7.3 Hz, 1H), 1.92 (d, J = 1.0 Hz, 3H), 1.83 (dq, J = 14.9, 7.5 Hz, 1H), 0.44 (t, J = 7.4 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 204.5, 147.7, 143.7, 142.1, 137.2, 133.5, 131.8, 128.8, 128.5, 128.4, 127.7, 127.2, 126.5, 126.3, 99.1, 70.9, 57.06, 42.3, 34.8, 15.7, 9.1. 93.5:6.5 e.r. as determined by HPLC (Chiralcel OD, 95:5 hexanes/i-PrOH, 1 mL/min), t_r (major) = 8.85 min, t_r (minor) = 8.18 min. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{23}H_{25}O_2$ 333.1849, found 333.1861.

(*R*,*E*)-1-(3-(Benzyloxy)allyl)-3-methyl-1-propylnaphthalen-2(1*H*)-one (3n). Colorless oil, (petroleum ether/EtOAc = 40:1−10:1); 45.1 mg, yield (65%); $[\alpha]_D^{20}$ = +4.15 (c 0.4, $CHCl_3$); ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.34−7.22 (m, 6H), 7.20−7.18 (m, 1H), 7.15−7.13 (m, 3H), 6.03 (d, J = 12.5 Hz, 1H), 4.48 (s, 2H), 4.26 (ddd, J = 12.5, 8.5, 7.1 Hz, 1H), 2.63 (dd, J = 13.4, 8.6 Hz, 1H), 2.39−2.32 (m, 1H), 2.17 (ddd, J = 12.9, 11.4, 4.8 Hz, 1H), 1.91 (d, J = 1.2 Hz, 3H), 1.76 (td, J = 12.4, 4.0 Hz, 1H), 0.88−0.82 (m, 1H), 0.75−0.67 (m, 4H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 204.36, 147.7, 144.0, 142.0, 137.2, 133.4, 131.6, 128.7, 128.4, 128.3, 127.7, 127.2, 126.4, 126.3, 99.1, 70.9, 56.5, 44.2, 42.5, 18.0, 15.7, 14.4. 94.5:5.5 e.r. as determined by HPLC (Chiralcel IC, 95:5 hexanes/i-PrOH, 1 mL/min), t_r (major) = 10.84 min, t_r (minor) = 11.77 min. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{24}H_{27}O_2$ 347.2006, found 347.2005.

(*R*,*E*)-1-(3-(Benzyloxy)allyl)-1-butyl-3-methylnaphthalen-2(1*H*)-one (3o). Colorless oil, (petroleum ether/EtOAc = 40:1−10:1); 44.0 mg, yield (61%); $[\alpha]_D^{20}$ = −13.60 (c 0.4, $CHCl_3$); ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.34−7.23 (m, 6H), 7.20 (d, J = 7.1 Hz, 1H), 7.15−7.13 (m, 3H), 6.03 (d, J = 12.5 Hz, 1H), 4.48 (s, 2H), 4.25 (ddd, J = 12.5, 8.5, 7.2 Hz, 1H), 2.62 (dd, J = 13.4, 8.6 Hz, 1H), 2.35 (dd, J = 13.3, 7.0 Hz, 1H), 2.19 (td, J = 12.5, 4.7 Hz, 1H), 1.91 (d, J = 1.0 Hz, 3H), 1.78 (td, J = 12.6, 4.2 Hz, 1H), 1.17−1.03 (m, 2H), 0.80−0.65 (m, 5H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 204.4, 147.7, 144.1, 142.0, 137.2, 133.4, 131.6, 128.7, 128.4, 128.4, 127.7, 127.2, 126.4, 126.2, 99.0, 70.9, 56.4, 42.7, 41.7, 26.8, 23.0, 15.7, 13.8. 94.5:5.5 e.r. as determined by HPLC (Chiralcel IC, 90:10 hexanes/i-PrOH, 1 mL/min), t_r (major) = 9.34 min, t_r (minor) = 10.26 min. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{25}H_{29}O_2$ 361.2162, found 361.2178.

(*R*,*E*)-1-(3-(Benzyloxy)allyl)-1-hexyl-3-methylnaphthalen-2(1*H*)-one (3p). Colorless oil, (petroleum ether/EtOAc = 40:1−8:1); 62.2 mg, yield (80%); $[\alpha]_D^{20}$ = −12.58 (c 0.4, $CHCl_3$); ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.34−7.22 (m, 6H), 7.19 (d, J = 6.8 Hz, 1H), 7.15−7.13 (m, 3H), 6.03 (d, J = 12.5 Hz, 1H), 4.48 (s, 2H), 4.26 (ddd, J = 12.5, 8.5, 7.2 Hz, 1H), 2.62 (dd, J = 13.4, 8.6 Hz, 1H), 2.35 (dd, J = 13.4, 7.0 Hz, 1H), 2.18 (td, J = 12.5, 4.8 Hz, 1H), 1.91 (d, J = 1.1 Hz, 3H), 1.78 (td, J = 12.6, 4.3 Hz, 1H), 1.16−1.03 (m, 6H), 0.78 (t, J = 7.2 Hz, 4H), 0.69 (ddd, J = 12.7, 8.6, 4.4 Hz, 1H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 204.4, 147.7, 144.1, 142.0, 137.2, 133.4, 131.6, 128.7, 128.4, 128.4, 127.7, 127.2, 126.4, 126.2, 99.1, 70.9, 56.4, 42.7, 41.9, 31.5, 29.6, 24.6, 22.5, 15.7, 14.0. 94.5:5.5 e.r. as determined by HPLC (Chiralcel AD, 95:5 hexanes/i-PrOH, 1 mL/min), t_r (major) = 7.88 min, t_r (minor) = 7.22 min. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{27}H_{33}O_2$ 389.2475, found 389.2470.

(*R*,*E*)-1-Allyl-1-(3-(benzyloxy)allyl)-3-methylnaphthalen-2(1*H*)-one (3q). Colorless oil, (petroleum ether/EtOAc = 40:1−10:1); 44.9 mg, yield (65%); $[\alpha]_D^{20}$ = −6.00 (c 0.4, $CHCl_3$); ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.34−7.33 (m, 2H), 7.30−7.23 (m, 4H), 7.19 (d, J = 7.5 Hz, 1H), 7.15−7.12 (m, 3H), 6.06 (d, J = 12.5 Hz, 1H), 5.25−5.16 (m, 1H), 4.84−4.69 (m, 2H), 4.49 (s, 2H), 4.28 (ddd, J = 12.5, 8.4, 7.2 Hz, 1H), 2.89 (dd, J = 13.6, 7.4 Hz, 1H), 2.68 (dd, J = 13.5, 8.5 Hz, 1H), 2.55 (dd, J = 13.5, 6.8 Hz, 1H), 2.44−2.37 (m, 1H), 1.90 (d, J = 1.1 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 203.5, 148.0, 143.2, 141.9, 137.1, 133.3, 133.2, 131.5, 128.7, 128.5, 128.4, 127.7, 127.2, 126.6, 117.6, 98.9, 70.9, 56.2, 45.6, 41.4, 15.7. 93.5:6.5 e.r. as determined by HPLC (Chiralcel AD, 99:1 hexanes/i-PrOH, 1 mL/min), t_r (major) = 9.64 min, t_r (minor) = 9.03 min. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{24}H_{24}O_2$ 345.1849, found 345.1862.

(*R*,*E*)-1-(3-(Benzyloxy)allyl)-3-ethyl-1-methylnaphthalen-2(1*H*)-one (3r). Colorless oil, (petroleum ether/EtOAc = 40:1−10:1); 50.6 mg, yield (76%); $[\alpha]_D^{20}$ = −88.66 (c 0.4, $CHCl_3$); ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.33−7.24 (m, 7H), 7.17 (d, J = 6.7 Hz, 2H), 7.13 (s, 1H), 6.08 (d, J = 12.5 Hz, 1H), 4.50 (s, 2H), 4.37−4.28 (m, 1H), 2.62 (dd, J = 13.5, 8.5 Hz, 1H), 2.42−2.34 (m, 3H), 1.43 (s, 3H), 1.11 (t, J = 7.5 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 203.9, 148.0, 145.0, 140.0, 138.0, 137.1, 130.4, 128.7, 128.4, 127.8, 127.3, 126.6, 126.4, 99.2, 70.2, 51.8, 42.2, 25.7, 22.4, 12.8. 93.5:6.5 e.r. as determined by HPLC (Chiralcel IB, 95:5 hexanes/i-PrOH, 1 mL/min), t_r (major) = 5.92 min, t_r (minor) = 5.47 min. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{23}H_{25}O_2$ 330.1849, found 333.1857.

(*R*,*E*)-3-Allyl-1-(3-(benzyloxy)allyl)-1-methylnaphthalen-2(1*H*)-one (3s). Colorless oil, (petroleum ether/EtOAc = 40:1−10:1); 61.4 mg, yield (89%); $[\alpha]_D^{20}$ = −91.00 (c 0.4, $CHCl_3$); ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.33−7.24 (m, 7H), 7.18−7.15 (m, 3H), 6.08 (d, J = 12.5 Hz, 1H), 5.89 (ddt, J = 16.9, 10.1, 6.8 Hz, 1H), 5.15−5.09 (m, 2H), 4.50 (s, 2H), 4.32 (dt, J = 12.6, 7.8 Hz, 1H), 3.12−3.10 (m, 2H), 2.63 (dd, J = 13.5, 8.5 Hz, 1H), 2.37 (dd, J = 13.5, 7.1 Hz, 1H), 1.43 (s, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 203.4, 148.1, 145.1, 141.5, 137.1, 135.5, 134.6, 130.2, 129.0, 128.9, 128.4, 127.8, 127.3, 126.7, 126.4, 116.8, 99.2, 70.9, 52.0, 42.1, 33.4,

https://dx.doi.org/10.1021/acs.joc.0c00582
J. Org. Chem. 2020, 85, 7896−7904

25.8. 93:8 e.r. as determined by HPLC (Chiralcel OD, 90:10 hexanes/*i*-PrOH, 1 mL/min), t_r (major) = 8.11 min, t_r (minor) = 7.23 min. HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{24}H_{25}O_2$ 345.1849, found 345.1854.

(*R*,*E*)-1-(3-(Benzyloxy)allyl)-1-methyl-3-phenylnaphthalen-2(1*H*)-one (3t). Colorless oil, (petroleum ether/EtOAc = 40:1−10:1); 57.2 mg, yield (75%); $[\alpha]_D^{20}$ = −32.95 (*c* 0.4, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 7.48−7.46 (m, 2H), 7.41−7.29 (m, 7H), 7.25−7.24 (m, 3H), 7.15 (dd, *J* = 7.0, 2.5 Hz, 2H), 6.15 (d, *J* = 12.5 Hz, 1H), 4.52 (d, *J* = 2.3 Hz, 2H), 4.44 (ddd, *J* = 12.5, 8.4, 7.3 Hz, 1H), 2.71 (dd, *J* = 13.9, 8.5 Hz, 1H), 2.45 (ddd, *J* = 13.6, 7.1, 0.9 Hz, 1H), 1.52 (s, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 202.6, 148.2, 145.5, 142.4, 137.0, 136.1, 135.5, 130.3, 129.7, 129.5, 128.8, 128.4, 128.2, 128.0, 127.8, 127.3, 126.9, 126.5, 99.2, 77.3, 77.1, 76.8, 71.0, 52.8, 42.4, 25.4. 92.5:7.5 e.r. as determined by HPLC (Chiralcel AD, 95:5 hexanes/*i*-PrOH, 1 mL/min), t_r (major) = 13.21 min, t_r (minor) = 12.24 min. HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{27}H_{25}O_2$ 381.1849, found 381.1857.

(*R*,*E*)-1-(3-(Benzyloxy)allyl)-1,3-dimethyl-6-phenylnaphthalen-2(1*H*)-one (3u). Colorless oil, (petroleum ether/EtOAc = 40:1−10:1); 56.1 mg, yield (71%); $[\alpha]_D^{20}$ = +2.15 (*c* 0.4, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 7.62−7.60 (m, 2H), 7.55 (dd, *J* = 8.1, 1.9 Hz, 1H), 7.46 (t, *J* = 7.6 Hz, 2H), 7.42 (d, *J* = 1.8 Hz, 1H), 7.38−7.36 (m, 2H), 7.29−7.24 (m, 3H), 7.22 (s, 1H), 7.18−7.16 (m, 2H), 6.11 (d, *J* = 12.5 Hz, 1H), 4.53 (s, 2H), 4.37 (ddd, *J* = 12.5, 8.4, 7.2 Hz, 1H), 2.65 (dd, *J* = 13.6, 8.6 Hz, 1H), 2.40 (ddd, *J* = 13.6, 7.1, 0.8 Hz, 1H), 1.96 (d, *J* = 1.1 Hz, 3H), 1.45 (s, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 204.2, 148.0, 144.0, 141.6, 140.3, 139.5, 137.1, 132.9, 130.8, 128.9, 128.4, 127.7, 127.6, 127.4, 127.2, 127.0, 127.0, 126.9, 99.4, 71.0, 51.7, 42.0, 26.0, 16.0. 91.5:8.5 e.r. as determined by HPLC (Chiralcel IA, 99:1 hexanes/*i*-PrOH, 1 mL/min), t_r (major) = 11.40 min, t_r (minor) = 12.27 min. HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{28}H_{27}O_2$ 395.2006, found 395.2001.

(*S*,*E*)-2-Allyl-2-(3-(benzyloxy)allyl)naphthalen-1(2*H*)-one (5a). Yellow oil, (petroleum ether/EtOAc = 40:1−8:1); 43.7 mg, yield (66%); $[\alpha]_D^{20}$ = −77.43 (*c* 0.4, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 8.01 (d, *J* = 7.4 Hz, 1H), 7.53 (td, *J* = 7.5, 1.2 Hz, 1H), 7.33 (t, *J* = 7.1 Hz, 1H), 7.29−7.17 (m, 6H), 6.66 (d, *J* = 9.8 Hz, 1H), 6.28 (d, *J* = 12.5 Hz, 1H), 6.04 (d, *J* = 9.8 Hz, 1H), 5.53 (ddt, *J* = 17.3, 10.1, 7.3 Hz, 1H), 4.99 (dd, *J* = 17.0, 1.4 Hz, 1H), 4.89 (dd, *J* = 10.1, 0.9 Hz, 1H), 4.62 (dt, *J* = 12.5, 7.9 Hz, 1H), 4.49 (s, 2H), 2.67 (dd, *J* = 13.5, 7.3 Hz, 1H), 2.50 (dd, *J* = 13.6, 8.2 Hz, 1H), 2.27 (dd, *J* = 13.5, 7.2 Hz, 1H), 2.19 (dd, *J* = 13.6, 7.9 Hz, 1H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 202.8, 148.3, 138.4, 137.9, 136.9, 134.3, 133.2, 129.8, 128.4, 127.8, 127.8, 127.4, 127.4, 126.6, 125.5, 118.0, 98.8, 71.0, 54.0, 43.4, 38.1. 91.5:8.5 e.r. as determined by HPLC (Chiralcel OD, 95:5 hexanes/*i*-PrOH, 1 mL/min), t_r (major) = 8.43 min, t_r (minor) = 7.50 min. HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{23}H_{23}O_2$ 331.1693, found 331.1712.

(*S*,*E*)-2-Allyl-2-(3-((4-ethylbenzyl)oxy)allyl)naphthalen-1(2*H*)-one (5b). Yellow oil, (petroleum ether/EtOAc = 40:1−8:1); 44.4 mg, yield (62%); $[\alpha]_D^{20}$ = −100.95 (*c* 0.4, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 8.01 (d, *J* = 7.8 Hz, 1H), 7.53 (t, *J* = 7.5 Hz, 1H), 7.33 (t, *J* = 7.6 Hz, 1H), 7.21 (d, *J* = 7.5 Hz, 1H), 7.11 (s, 4H), 6.66 (d, *J* = 9.8 Hz, 1H), 6.27 (d, *J* = 12.4 Hz, 1H), 6.04 (d, *J* = 9.8 Hz, 1H), 5.56−5.50 (m, 1H), 4.99 (d, *J* = 17.0 Hz, 1H), 4.90−4.88 (m, 1H), 4.62 (dt, *J* = 12.5, 7.9 Hz, 1H), 4.46 (s, 2H), 2.69−2.52 (m, 3H), 2.52−2.46 (m, 1H), 2.28 (dd, *J* = 13.5, 7.2 Hz, 1H), 2.19 (dd, *J* = 13.6, 7.5 Hz, 1H), 1.20 (t, *J* = 7.6 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 202.8, 148.4, 144.0, 138.47, 137.9, 134.3, 134.16, 133.3, 129.9, 127.9, 127.8, 127.8, 127.7, 127.6, 127.4, 126.6, 125.5, 117.9, 98.7, 71.0, 54.0, 43.3, 38.2, 28.6, 15.6. 92:8 e.r. as determined by HPLC (Chiralcel IA, 99:1 hexanes/*i*-PrOH, 1 mL/min), t_r (major) = 11.16 min, t_r (minor) = 10.30 min. HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{25}H_{27}O_2$ 359.2006, found 359.1998.

(*S*,*E*)-2-Allyl-2-(3-((3,5-dimethoxybenzyl)oxy)allyl)-naphthalen-1(2*H*)-one (5c). Yellow oil, (petroleum ether/EtOAc = 40:1−8:1); 55.4 mg, yield (71%); $[\alpha]_D^{20}$ = −82.73 (*c* 0.4, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 8.07−7.94 (m, 1H), 7.53 (td, *J* = 7.5, 1.3 Hz, 1H), 7.33 (td, *J* = 7.6, 1.0 Hz, 1H), 7.21 (d, *J* = 7.4 Hz, 1H), 6.66 (d, *J* = 9.8 Hz, 1H), 6.41−6.31 (m, 3H), 6.26 (d, *J* = 12.5 Hz, 1H), 6.04 (d, *J* = 9.8 Hz, 1H), 5.57−5.49 (m, 1H), 5.01−4.97 (m, 1H), 4.91−4.85 (m, 1H), 4.61 (dt, *J* = 12.5, 7.8 Hz, 1H), 4.44 (s, 2H), 3.74 (d, *J* = 1.7 Hz, 6H), 2.67 (dd, *J* = 13.5, 7.3 Hz, 1H), 2.50 (dd, *J* = 14.0, 7.8 Hz, 1H), 2.28 (dd, *J* = 13.5, 7.2 Hz, 1H), 2.18 (ddd, *J* = 13.6, 7.5, 0.9 Hz, 1H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 202.7, 160.8, 148.1, 139.3, 138.4, 137.9, 134.3, 133.2, 129.8, 127.8, 127.4, 126.6, 125.5, 117.9, 105.1, 100.0, 99.0, 71.0, 55.3, 53.9, 43.3, 38.0. 92:8 e.r. as determined by HPLC (Chiralcel AD, 90:10 hexanes/*i*-PrOH, 1 mL/min), t_r (major) = 11.41 min, t_r (minor) = 8.96 min. HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{25}H_{27}O_4$ 391.1904, found 391.1922.

(*S*,*E*)-2-Allyl-2-(3-((4-fluorobenzyl)oxy)allyl)naphthalen-1(2*H*)-one (5d). Yellow oil, (petroleum ether/EtOAc = 40:1−8:1); 45.2 mg, yield (65%); $[\alpha]_D^{20}$ = −92.33 (*c* 0.4, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 8.00 (d, *J* = 7.8 Hz, 1H), 7.54 (t, *J* = 7.5 Hz, 1H), 7.33 (t, *J* = 7.6 Hz, 1H), 7.21 (d, *J* = 7.6 Hz, 1H), 7.13 (dd, *J* = 7.2, 5.8 Hz, 2H), 6.94−6.91 (m, 2H), 6.65 (d, *J* = 9.8 Hz, 1H), 6.25 (d, *J* = 12.5 Hz, 1H), 6.03 (dd, *J* = 9.8, 0.9 Hz, 1H), 5.55−5.50 (m, 1H), 4.99 (d, *J* = 17.0 Hz, 1H), 4.89 (d, *J* = 10.1 Hz, 1H), 4.59 (dt, *J* = 12.5, 7.8 Hz, 1H), 4.46 (s, 2H), 2.66 (dd, *J* = 13.4, 7.2 Hz, 1H), 2.52 (dd, *J* = 13.6, 8.3 Hz, 1H), 2.28 (dd, *J* = 13.5, 7.2 Hz, 1H), 2.18 (dd, *J* = 13.6, 7.4 Hz, 1H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 202.7, 162.3 (d, *J* = 244.4 Hz), 148.0, 138.4, 137.8, 134.3, 133.1, 132.7 (d, *J* = 3.1 Hz), 129.1 (d, *J* = 8.2 Hz), 129.1, 127.8, 127.4, 126.6, 125.6, 118.0, 115.3 (d, *J* = 21.3 Hz), 99.1, 70.2, 54.0, 43.5, 38.0. 91.5:8.5 e.r. as determined by HPLC (Chiralcel AD, 99:1 hexanes/*i*-PrOH, 1 mL/min), t_r (major) = 21.04 min, t_r (minor) = 17.95 min. HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{23}H_{22}FO_2$ 349.1598, found 349.1591.

(*S*,*E*)-2-Allyl-2-(3-(benzyloxy)allyl)-4-methoxynaphthalen-1(2*H*)-one (5e). Yellow oil, (petroleum ether/EtOAc = 40:1−8:1); 56.9 mg, yield (79%); $[\alpha]_D^{20}$ = −278.90 (*c* 0.4, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 8.00 (dd, *J* = 7.8, 1.0 Hz, 1H), 7.73 (dd, *J* = 7.9, 0.7 Hz, 1H), 7.59 (td, *J* = 7.6, 1.4 Hz, 1H), 7.38 (td, *J* = 7.6, 1.2 Hz, 1H), 7.27−7.18 (m, 5H), 6.26 (d, *J* = 12.5 Hz, 1H), 5.60−5.51 (m, 1H), 5.00−4.97 (m, 1H), 4.95 (s, 1H), 4.91−4.88 (m, 1H), 4.65 (dt, *J* = 12.5, 8.0 Hz, 1H), 4.51 (s, 2H), 3.75 (s, 3H), 2.65 (dd, *J* = 13.4, 7.4 Hz, 1H), 2.48 (ddd, *J* = 13.5, 8.3, 0.7 Hz, 1H), 2.27 (dd, *J* = 13.4, 7.2 Hz, 1H), 2.18 (ddd, *J* = 13.5, 7.4, 1.0 Hz, 1H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 202.2, 149.5, 148.2, 137.0, 135.9, 134.1, 133.6, 130.0, 128.4, 128.2, 127.8, 127.4, 126.6, 122.4, 117.9, 104.5, 99.3, 71.1, 54.9, 52.8, 44.4, 39.1. 94:6 e.r. as determined by HPLC (Chiralcel OD, 90:10 hexanes/*i*-PrOH, 1 mL/min), t_r (major) = 9.44 min, t_r (minor) = 5.12 min. HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{24}H_{25}O_3$ 361.1798, found 361.18.

(*S*,*E*)-2-Allyl-2-(3-(benzyloxy)allyl)-4-ethoxynaphthalen-1(2*H*)-one (5f). Yellow oil, (petroleum ether/EtOAc = 20:1−4:1); 59.1 mg, yield (79%); $[\alpha]_D^{20}$ = −153.90 (*c* 0.4, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 7.99 (d, *J* = 7.8 Hz, 1H), 7.77 (d, *J* = 7.9 Hz, 1H), 7.61−7.58 (m, 1H), 7.37 (t, *J* = 7.5 Hz, 1H), 7.28−7.23 (m, 3H), 7.18 (d, *J* = 7.0 Hz, 2H), 6.26 (d, *J* = 12.5 Hz, 1H), 5.56 (ddt, *J* = 17.2, 10.1, 7.2 Hz, 1H), 4.99−4.88 (m, 3H), 4.65 (dt, *J* = 12.6, 7.8 Hz, 1H), 4.51 (s, 2H), 3.93 (tq, *J* = 6.3, 2.0 Hz, 2H), 2.64 (dd, *J* = 13.4, 7.4 Hz, 1H), 2.47 (dd, *J* = 13.5, 8.3 Hz, 1H), 2.25 (dd, *J* = 13.4, 7.2 Hz, 1H), 2.17 (dd, *J* = 13.5, 7.4 Hz, 1H), 1.45 (t, *J* = 6.9 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 202.3, 148.6, 148.2, 137.1, 136.1, 134.0, 133.7, 130.1, 128.4, 128.1, 127.8, 127.37, 126.5, 122.5, 117.9, 105.2, 99.4, 77.3, 77.1, 76.8, 71.1, 63.0, 52.8, 44.4, 39.1, 14.6. 93:7 e.r. as determined by HPLC (Chiralcel OD, 95:5 hexanes/*i*-PrOH, 1 mL/min), t_r (major) = 6.60 min, t_r (minor) = 5.83 min. HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{25}H_{27}O_3$ 375.1955, found 375.197.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.0c00582.

Copies of ^1H, ^{19}F, and ^{13}C NMR spectra of all new compounds (PDF)

https://dx.doi.org/10.1021/acs.joc.0c00582
J. Org. Chem. 2020, 85, 7896−7904

AUTHOR INFORMATION

Corresponding Author

Xiaofei Zeng − *College of Material Chemistry and Chemical Engineering, Hangzhou Normal University, Hangzhou 310036, China;* orcid.org/0000-0003-4222-1365; Email: chemzxf@hznu.edu.cn

Authors

Jinjin Hu − *College of Material Chemistry and Chemical Engineering, Hangzhou Normal University, Hangzhou 310036, China*

Shulei Pan − *College of Material Chemistry and Chemical Engineering, Hangzhou Normal University, Hangzhou 310036, China*

Shuai Zhu − *Shenzhen Grubbs Institute and Department of Chemistry, Southern University of Science and Technology, Shenzhen 518055, China*

Peiyuan Yu − *Shenzhen Grubbs Institute and Department of Chemistry, Southern University of Science and Technology, Shenzhen 518055, China;* orcid.org/0000-0002-4367-6866

Ruigang Xu − *College of Material Chemistry and Chemical Engineering, Hangzhou Normal University, Hangzhou 310036, China*

Guofu Zhong − *College of Material Chemistry and Chemical Engineering, Hangzhou Normal University, Hangzhou 310036, China;* orcid.org/0000-0001-9497-9069

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.joc.0c00582

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We gratefully acknowledge the Natural Science Foundation of China (no. 21672048), the Natural Science Foundation of Zhejiang Province (no. LY18B020015), and Hangzhou Normal University for financial support. X.Z. acknowledges a Xihu Scholar award from Hangzhou City, and G.Z. acknowledges a Qianjiang Scholar from Zhejiang Province, China.

REFERENCES

(1) (a) Majetich, G.; Grove, J. L. Total Synthesis of (+)-19-Deoxyicetexone, (−)-Icetexone, and (+)-5-Epi-icetexone. *Org. Lett.* **2009**, *11*, 2904−2907. (b) Myers, A. G.; Herzon, S. B. Identification of a Novel Michael Acceptor Group for the Reversible Addition of Oxygen- and Sulfur-Based Nucleophiles. Synthesis and Reactivity of the 3-Alkylidene-3*H*-indole 1-Oxide Function of Avrainvillamide. *J. Am. Chem. Soc.* **2003**, *125*, 12080−12081. (c) Xu, Z.; Wang, Q.; Zhu, J. Total Syntheses of (−)-Mersicarpine, (−)-Scholarisine G, (+)-Melodinine E, (−)-Leuconoxine, (−)-Leuconolam, (−)-Leuconodine A, (+)-Leuconodine F, and (−)-Leuconodine C: Self-Induced Diastereomeric Anisochronism (SIDA) Phenomenon for Scholarisine G and Leuconodines A and C. *J. Am. Chem. Soc.* **2015**, *137*, 6712−6724.

(2) (a) Amand, S.; Vallet, M.; Guedon, L.; Genta-Jouve, G.; Wien, F.; Mann, S.; Dupont, J.; Prado, S.; Nay, B. A Reactive Eremophilane and Its Antibacterial 2(1*H*)-Naphthalenone Rearrangement Product, Witnesses of a Microbial Chemical Warfare. *Org. Lett.* **2017**, *19*, 4038−4041. (b) Hutchinson, D. K. Anti-infective Agents. US Patent US200,501,073,64. 2005. (c) Decosterd, L. A.; Stoeckli-Evans, H.; Chapuis, J. C.; Sordat, B.; Hostettmann, K. New Cell Growth-Inhibitory Cyclohexadienone Derivatives from Hypericum calycinum L. *Theor. Chim. Acta* **1989**, *72*, 1833−1845.

(3) (a) Hart, H.; Lange, R. M. Direct Oxidation of Durene to a Cyclohexadienone. Chemistry and Photolysis[1]. *J. Org. Chem.* **1966**, *31*, 3776−3779. (b) Coleman, R. S.; Guernon, J. M.; Roland, J. T. Synthesis of the Spirocyclic Cyclohexadienone Ring System of the Schiarisanrins. *Org. Lett.* **2000**, *2*, 277−280. (c) Jackson, S. R.; Johnson, M. G.; Mikami, M.; Shiokawa, S.; Carreira, E. M. Rearrangement of a Tricyclic 2,5-Cyclohexadienone: Towards a General Synthetic Route to the Daphnanes and (+)- Resiniferatoxin. *Angew. Chem., Int. Ed.* **2001**, *40*, 2694−2697. (d) Liu, P.; Fukui, Y.; Tian, P.; He, Z.-T.; Sun, C.-Y.; Wu, N.-Y.; Lin, G.-Q. Cu-Catalyzed Asymmetric Borylative Cyclization of Cyclohexadienone-Containing 1,6-Enynes. *J. Am. Chem. Soc.* **2013**, *135*, 11700−11703. (e) Reddy, R. R.; Gudup, S. S.; Ghorai, P. Organocatalytic, Enantioselective Synthesis of Cyclohexadienone Containing Hindered Spirocyclic Ethers through an Oxidative Dearomatization/Oxa-Michael Addition Sequence. *Angew. Chem., Int. Ed.* **2016**, *55*, 15115−15119.

(4) (a) Zhuo, C.-X.; Zhang, W.; You, S.-L. Catalytic Asymmetric Dearomatization Reactions. *Angew. Chem., Int. Ed.* **2012**, *51*, 12662−12686. (b) Zhuo, C.-X.; Zheng, C.; You, S.-L. Transition-Metal-Catalyzed Asymmetric Allylic Dearomatization Reactions. *Acc. Chem. Res.* **2014**, *47*, 2558−2573. (c) Zheng, C.; You, S.-L. Catalytic Asymmetric Dearomatization by Transition-Metal Catalysis: A Method for Transformations of Aromatic Compounds. *Chem* **2016**, *1*, 830−857.

(5) (a) Magdziak, D.; Meek, S. J.; Pettus, T. R. R. Cyclohexadienone Ketals and Quinols: Four Building Blocks Potentially Useful for Enantioselective Synthesis. *Chem. Rev.* **2004**, *104*, 1383−1430. (b) Wu, W.-T.; Zhang, L.; You, S.-L. Catalytic asymmetric dearomatization (CADA) reactions of phenol and aniline derivatives. *Chem. Soc. Rev.* **2016**, *45*, 1570−1580. (c) Sun, W.-S.; Li, G.-F.; Hong, L.; Wang, R. Asymmetric Dearomatization of Phenols. *Org. Biomol. Chem.* **2016**, *14*, 2164−2176.

(6) (a) Tu, H.-F.; Zheng, C.; Xu, R.-Q.; Liu, X.-J.; You, S.-L. Iridium-Catalyzed Intermolecular Asymmetric Dearomatization of *β*-Naphthols with Allyl Alcohols or Allyl Ethers. *Angew. Chem., Int. Ed.* **2017**, *56*, 3237−3241. (b) Shen, D.; Chen, Q.; Yan, P.; Zeng, X.; Zhong, G. Enantioselective Dearomatization of Naphthol Derivatives with Allylic Alcohols by Cooperative Iridium and Brønsted Acid Catalysis. *Angew. Chem., Int. Ed.* **2017**, *56*, 3242−3246. (c) Tang, S.-B.; Tu, H.-F.; Zhang, X.; You, S.-L. Rhodium-Catalyzed Asymmetric Allylic Dearomatization of *β*-Naphthols: Enantioselective Control of Prochiral Nucleophiles. *Org. Lett.* **2019**, *21*, 6130−6134.

(7) (a) Nemoto, T.; Ishige, Y.; Yoshida, M.; Kohno, Y.; Kanematsu, M.; Hamada, Y. Novel Method for Synthesizing Spiro[4.5]-cyclohexadienones through a Pd-Catalyzed Intramolecular ipso-Friedel−Crafts Allylic Alkylation of Phenols. *Org. Lett.* **2010**, *12*, 5020−5023. (b) Yoshida, M.; Nemoto, T.; Zhao, Z.; Ishige, Y.; Hamada, Y. Enantioselective construction of all-carbon quaternary spirocenters through a Pd-catalyzed asymmetric intramolecular ipso-Friedel−Crafts allylic alkylation of phenols. *Tetrahedron: Asymmetry* **2012**, *23*, 859−866. (c) Zhuo, C.-X.; You, S.-L. Palladium-Catalyzed Intermolecular Allylic Dearomatization Reaction of *α*-Substituted *β*-Naphthol Derivatives: Scope and Mechanistic Investigation. *Adv. Synth. Catal.* **2014**, *356*, 2020−2028. (d) Zhuo, C.-X.; You, S.-L. Palladium-Catalyzed Intermolecular Asymmetric Allylic Dearomatization Reaction of Naphthol Derivatives. *Angew. Chem., Int. Ed.* **2013**, *52*, 10056−10059. (e) Cheng, Q.; Wang, Y.; You, S.-L. Chemo-, Diastereo-, and Enantioselective Iridium-Catalyzed Allylic Intramolecular Dearomatization Reaction of Naphthol Derivatives. *Angew. Chem., Int. Ed.* **2016**, *55*, 3496−3499. (f) Yao, L.; Ishihara, K. Enantioselective [1,3] O-to-C Rearrangement: Dearomatization of Alkyl 2-Allyloxy/Benzyloxy1/3-Naphthoates Catalyzed by a Chiral p−Cu(II) Complex. *Chem. Sci.* **2019**, *10*, 2259−2263.

(8) (a) Fang, X.; Zeng, Y.; Li, Q.; Wu, Z.; Yao, H.; Lin, A. Redox-Neutral Atom-Economic Pd(0)-Catalyzed Dearomatization of *β*-Naphthols with Alkynes toward Naphthalenones. *Org. Lett.* **2018**, *20*, 2530−2533. (b) Fang, X.; Li, Q.; Shi, R.; Yao, H.; Lin, A. Controllable Pd-Catalyzed Allylation of Indoles with Skipped Enynes: Divergent

Synthesis of Indolenines and *N*-Allylindoles. *Org. Lett.* **2018**, *20*, 6084−6088.
(9) Jin, M.; Ren, W.; Qian, D.-W.; Yang, S.-D. Direct Allylic C(sp3)-H Alkylation with 2-Naphthols via Cooperative Palladium and Copper Catalysis: Construction of Cyclohexadienones with Quaternary Carbon Centers. *Org. Lett.* **2018**, *20*, 7015−7019.
(10) An, J.; Lombardi, L.; Grilli, S.; Bandini, M. PPh_3AuTFA Catalyzed in the Dearomatization of 2-Naphthols with Allenamides. *Org. Lett.* **2018**, *20*, 7380−7383.
(11) Yang, B.; Zhai, X.; Feng, S.; Hu, D.; Deng, Y.; Shao, Z. Organocatalyzed Intermolecular Asymmetric Allylic Dearomatization of Both α- and β-Naphthols. *Org. Lett.* **2019**, *21*, 330−334.
(12) Trost, B. M.; Jäkel, C.; Plietker, B. Palladium-Catalyzed Asymmetric Addition of Pronucleophiles to Allenes. *J. Am. Chem. Soc.* **2003**, *125*, 4438−4439.
(13) For a review, see: (a) Brasholz, M.; Reissig, H.-U.; Zimmer, R. Sugars, Alkaloids, and Heteroaromatics: Exploring Heterocyclic Chemistry with Alkoxyallenes. *Acc. Chem. Res.* **2009**, *42*, 45−56.
(14) For selected examples, see: (a) Trost, B. M.; Simas, A. B. C.; Plietker, B.; Jäkel, C.; Xie, J. Enantioselective Palladium-Catalyzed Addition of 1,3-Dicarbonyl Compounds to an Allene Derivative. *Chem. − Eur. J.* **2005**, *11*, 7075−7082. (b) Trost, B. M.; Xie, J. Palladium-Catalyzed Asymmetric Ring Expansion of Allenylcyclobutanols: An Asymmetric Wagner−Meerwein Shift. *J. Am. Chem. Soc.* **2006**, *128*, 6044−6045. (c) Trost, B. M.; Xie, J. Palladium-Catalyzed Diastereo-and Enantioselective Wagner−Meerwein Shift: Control of Absolute Stereochemistry in the C−C Bond Migration Event. *J. Am. Chem. Soc.* **2008**, *130*, 6231−6242. (d) Trost, B. M.; Xie, J.; Sieber, J. D. The Palladium Catalyzed Asymmetric Addition of Oxindoles and Allenes: An Atom-Economical Versatile Method for the Construction of Chiral Indole Alkaloids. *J. Am. Chem. Soc.* **2011**, *133*, 20611−20622.
(15) For selected examples, see: (a) Kim, H.; Lim, W.; Im, D.; Kim, D. G.; Rhee, Y. H. Synthetic Strategy for Cyclic Amines: A Stereodefined Cyclic *N,O*-Acetal as a Stereocontrol and Diversity-Generating Element. *Angew. Chem., Int. Ed.* **2012**, *51*, 12055−12058. (b) Kim, H.; Rhee, Y. H. Stereodefined *N,O*-Acetals: Pd-Catalyzed Synthesis from Homopropargylic Amines and Utility in the Flexible Synthesis of 2,6-Substituted Piperidines. *J. Am. Chem. Soc.* **2012**, *134*, 4011−4014. (c) Kinderman, S. S.; Wekking, M. M. T.; van Maarseveen, J. H.; Schoemaker, H. E.; Hiemstra, H.; Rutjes, F. P. J. T. Catalytic *N*-Sulfonyliminium Ion-Mediated Cyclizations to α-Vinyl-Substituted Isoquinolines and β-Carbolines and Applications in Metathesis. *J. Org. Chem.* **2005**, *70*, 5519−5527. (d) Lim, W.; Kim, J.; Rhee, Y. H. Pd-Catalyzed Asymmetric Intermolecular Hydroalkoxylation of Allene: An Entry to Cyclic Acetals with Activating Group-Free and Flexible Anomeric Control. *J. Am. Chem. Soc.* **2014**, *136*, 13618−13621. (e) Renaud, P.; Beaufils, F.; Feray, L.; Schenk, K. Diastereoselective Radical-Mediated Hydrogen-Atom Abstraction. *Angew. Chem., Int. Ed.* **2003**, *42*, 4230−4233. (f) Jiang, L.; Jia, T.; Wang, M.; Liao, J.; Cao, P. Pd-Catalyzed Enantioselective Hydroalkoxylation of Alkoxyallenes with Phenol for Construction of Acyclic O,O-Acetals. *Org. Lett.* **2015**, *17*, 1070−1073. (g) Zhou, H.; Wei, Z.; Zhang, J.; Yang, H.; Xia, C.; Jiang, G. From Palladium to Brønsted Acid Catalysis: Highly Enantioselective Regiodivergent Addition of Alkoxyallenes to Pyrazolones. *Angew. Chem. Int. Ed.* **2017**, *56*, 1077−1081.
(16) (a) Pan, S.; Wu, B.; Hu, J.; Xu, R.; Jiang, M.; Zeng, X.; Zhong, G. Palladium-Catalyzed Allylic Substitution Reaction of Benzothiazolylacetamide with Allylic Alcohols in Water. *J. Org. Chem.* **2019**, *84*, 10111. (b) Yan, P.; Pan, S.; Hu, J.; Lu, L.; Zeng, X.; Zhong, G. Palladium Catalyzed Controllable Mono- or Di-Allylic Substitution Reaction of Benzothiazolylacetate with Allylic Alcohols. *Adv. Synth. Catal.* **2019**, *361*, 1322.
(17) Liu, W.-B.; Zheng, C.; Zhuo, C.-X.; Dai, L.-X.; You, S.-L. Iridium-Catalyzed Allylic Alkylation Reaction with N-Aryl Phosphoramidite Ligands: Scope and Mechanistic Studies. *J. Am. Chem. Soc.* **2012**, *134*, 4812.
(18) (a) Di Bugno, C.; Pasquali, M.; Leoni, P.; Sabatino, P.; Braga, D. Oxidative addition of phenols to bis(tricyclohexylphosphine)-palladium. Synthesis and Structural Characterization of *trans*-$[Pd(PCy_3)_2(H)(OC_6H_5)].C_6H_5OH$ (**1**) and *trans*-$[Pd(PCy_3)_2(H)(OC_6F_5)].C_6F_5OH$ (**2**). *Inorg. Chem.* **1989**, *28*, 1390. (b) Perez, P. J.; Calabrese, J. C.; Bunel, E. E. Synthesis, Characterization, and Reactivity of $[((iPr)_2P(CH_2)_3P(iPr)_2)(PCy_3)PdH][OR]$. *Organometallics* **2001**, *20*, 337. (c) Grushin, V. V. Hydrido Complexes of Palladium. *Chem. Rev.* **1996**, *96*, 2011.
(19) (a) Hamilton, J. Y.; Sarlah, D.; Carreira, E. M. Iridium-Catalyzed Enantioselective Allylic Vinylation with Potassium Alkenyltrifluoroborates. *Org. Synth.* **2015**, *92*, 1−12. (b) Wu, Q.; Shao, P.-L.; He, Y. Synthesis of 1,4,5,6-tetrahydropyridazines and pyridazines via transition-metal-free (4 + 2) cycloaddition of alkoxyallenes with 1,2-diaza-1,3-dienes. *RSC Adv* **2019**, *9*, 21507−21512. (c) Schünemann, K.; Furkert, D. P.; Connelly, S.; Fraser, J. D.; Sperry, J.; Brimble, M. A. Synthesis and Biological Evaluation of 7-Deoxy Analogues of the Human Rhinovirus 3C Protease Inhibitor Thysanone. *Eur. J. Org. Chem.* **2014**, 122−128.

https://dx.doi.org/10.1021/acs.joc.0c00582
J. Org. Chem. 2020, 85, 7896−7904